STARALEX EDUCATIONAL COMPLEX

BASIC EDUCATION CERTIFICATE EXAMINATION (BECE)

MATHEMATICS — MOCK EXAMINATION SAMPLE PAPER

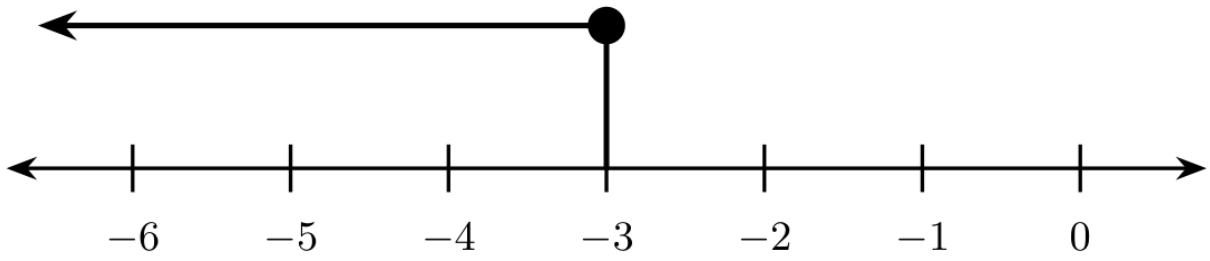
Time Allowed: 1 Hour | Paper 1 (Objective Test)

INSTRUCTIONS TO CANDIDATES

1. Use a soft pencil (type B or 2B) for answering all questions.
2. Read each question carefully before choosing your answer.
3. Each question is followed by four options lettered A to D. Find the correct option for each question and record your choice clearly.
4. All rough work must be done on scratch paper and not on this question paper.

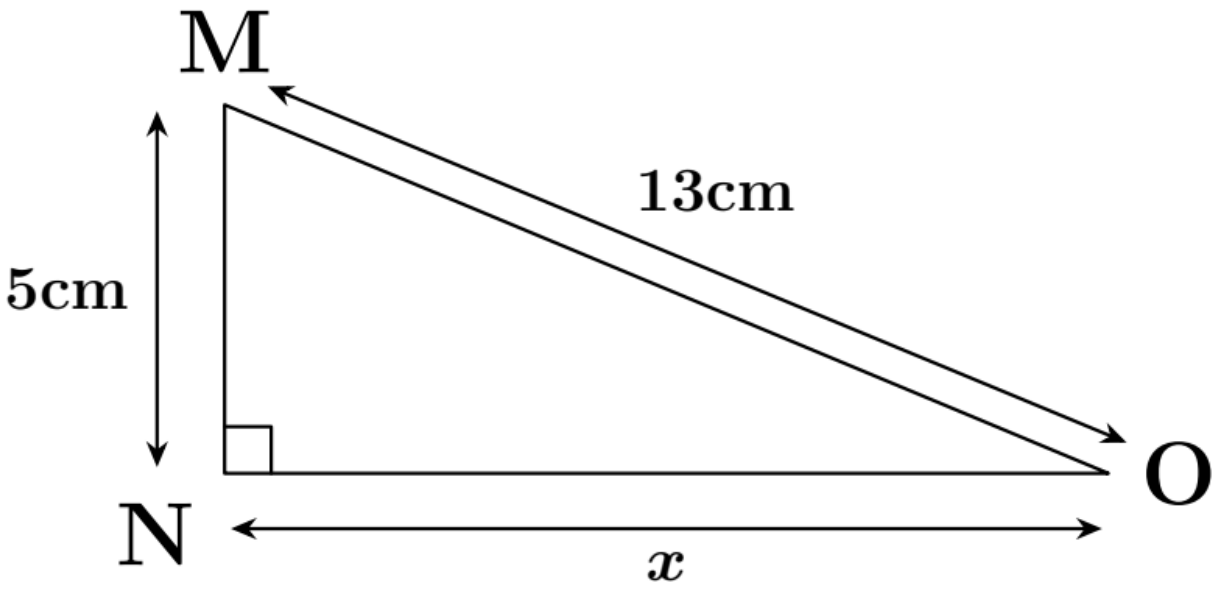
SECTION A: OBJECTIVE DRILLS

1. The number line below shows a mathematical solution set. Which of the following inequalities is represented by the diagram? (see Figure 1)



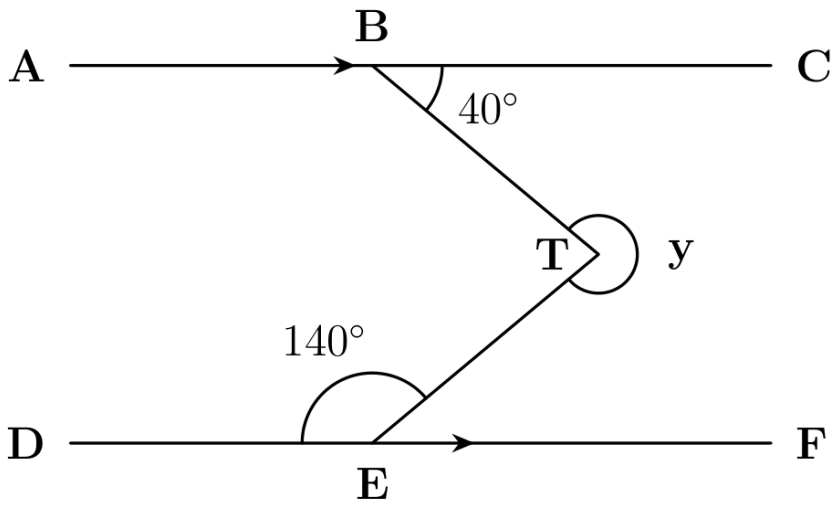
A. $x < -3$
B. $x \leq -3$
C. $x > -3$
D. $x \geq -3$

2. In the diagram below, MNO is a right-angled triangle where |MO| = 13 cm and |MN| = 5 cm. What is the value of x? (see Figure 2)



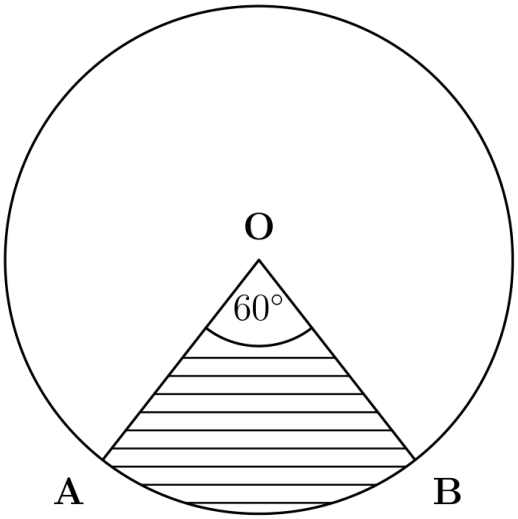
A. 3 cm
B. 4 cm
C. 8 cm
D. 12 cm

3. In the diagram below, line AC is parallel to line DF (AC // DF). Angle CBT is 40° and angle DET is 140°. Find the value of the reflex angle marked y. (see Figure 3)



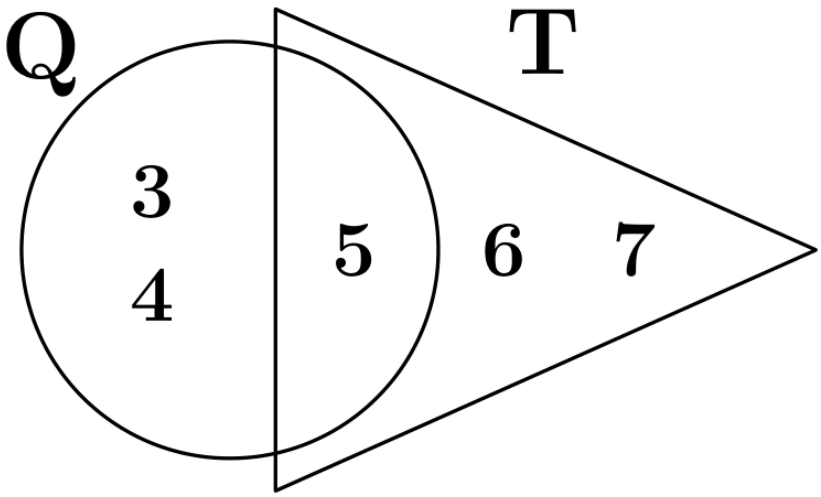
A. 320°
B. 280°
C. 220°
D. 100°

4. In the diagram below, O is the centre of the circle and r is its radius. Calculate the area of the shaded minor sector. (see Figure 4)



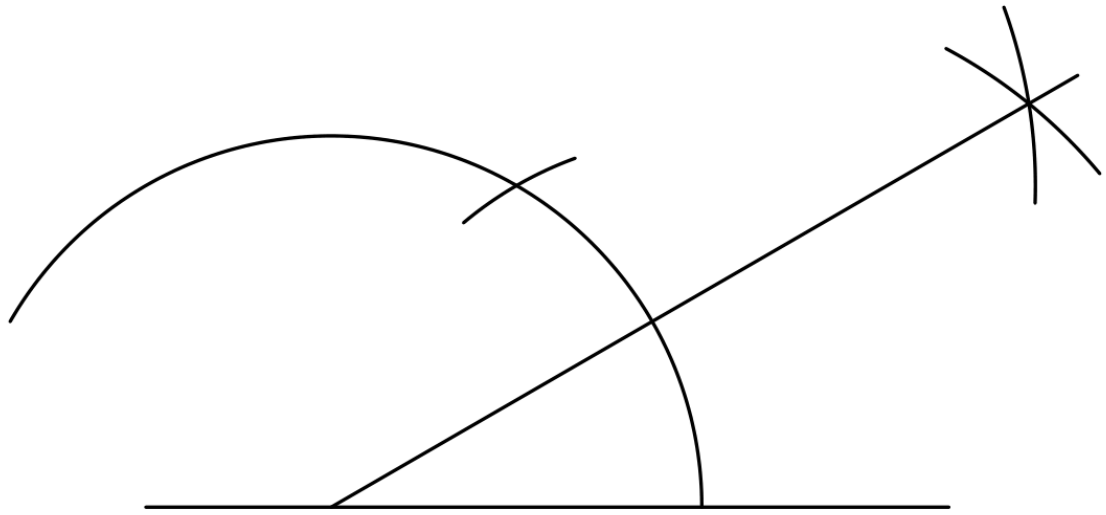
A. $\frac{1}{6}\pi r^2$
B. $\frac{1}{5}\pi r^2$
C. $\frac{1}{3}\pi r^2$
D. $\frac{5}{6}\pi r^2$

5. In the diagram below, Q is the set of numbers inside the circle and T is the set of numbers inside the triangle. Determine the union set Q ∪ T. (see Figure 5)



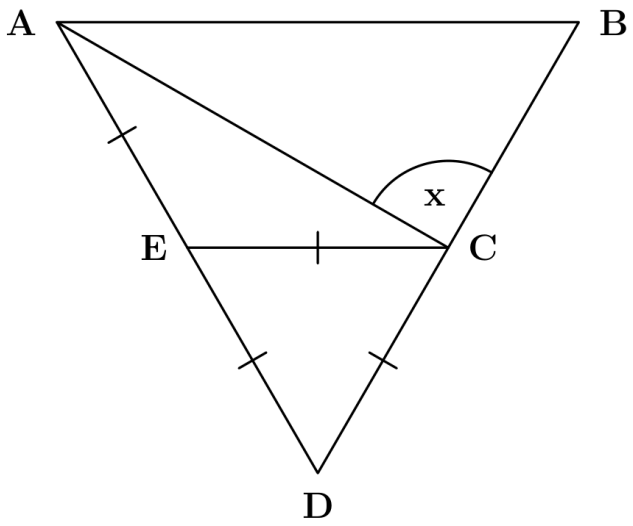
A. {5}
B. {6, 7}
C. {3, 4, 5}
D. {3, 4, 5, 6, 7}

6. Which of the following statements best describes the compass and straightedge construction shown in the diagram below? (see Figure 6)



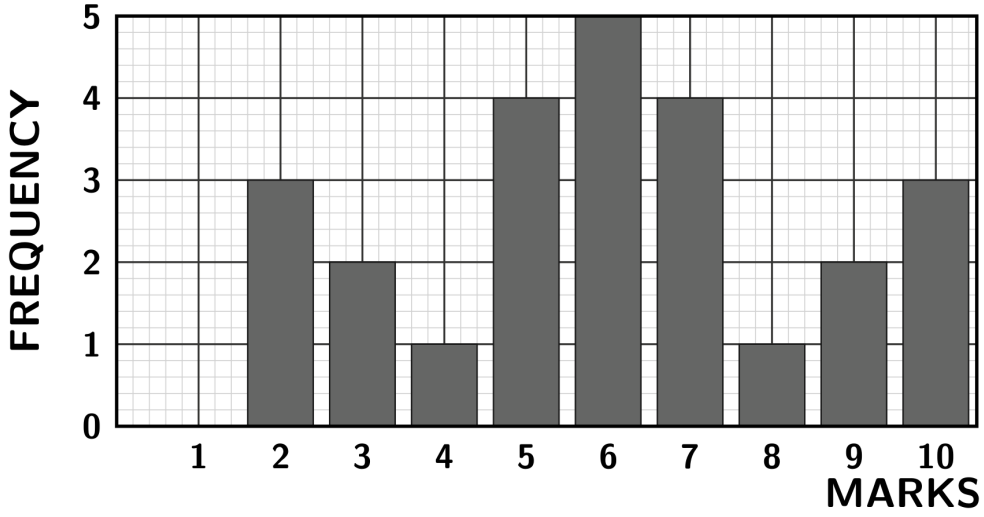
A. Constructing 30°
B. Constructing 60°
C. Constructing 120°
D. Constructing 135°

7. In the diagram below, |AE| = |ED| = |DC| = |CE|. Calculate the size of the angle marked x. (see Figure 7)



A. 120°
B. 90°
C. 60°
D. 30°

8. The frequency bar chart below shows the distribution of marks scored by pupils in a mathematics test. How many pupils took the test? (see Figure 8)



A. 20 pupils
B. 25 pupils
C. 30 pupils
D. 35 pupils

SECTION B: MODEL MARKING SCHEME & STEP-BY-STEP SOLUTIONS

Question 1

- Answer: B — $x \leq -3$
- Working & Reason: The arrow originates at −3 and points to the left towards smaller negative values, indicating a "less than" relation. The solid circle at −3 confirms that −3 is included in the solution set (≤). Thus, the inequality is $x \leq -3$.

Question 2

- Answer: D — 12 cm
- Working & Reason: By Pythagoras theorem in right-angled triangle MNO:

$|MO|^2 = |MN|^2 + |NO|^2$ $13^2 = 5^2 + x^2$ $169 = 25 + x^2$ $x^2 = 169 - 25 = 144$ $x = \sqrt{144} = 12$ cm.

Question 3

- Answer: B — 280°
- Working & Reason: Draw an auxiliary line through T parallel to AC and DF.

Upper alternate angle = ∠CBT = 40°. Lower alternate angle = 180° − ∠DET = 180° − 140° = 40°. Total interior angle ∠BTE = 40° + 40° = 80°. Reflex angle y = 360° − 80° = 280°.

Question 4

- Answer: A — $\frac{1}{6}\pi r^2$
- Working & Reason: The area of a circle sector is given by:

Area = $(\theta / 360°) \times \pi r^2$ Here $\theta = 60°$. Area = $(60° / 360°) \times \pi r^2 = \frac{1}{6}\pi r^2$.

Question 5

- Answer: D — {3, 4, 5, 6, 7}
- Working & Reason: Set Q contains all elements in the circle: Q = {3, 4, 5}.

Set T contains all elements in the triangle: T = {5, 6, 7}. The union Q ∪ T combines all unique elements from both sets: Q ∪ T = {3, 4, 5, 6, 7}.

Question 6

- Answer: A — Constructing 30°
- Working & Reason: The construction first establishes a 60° arc intersection using equal compass radii from the baseline. Next, equal arcs are struck from the baseline intersection and the 60° mark to bisect the 60° angle. A ray drawn through this bisection yields half of 60°, which is exactly 30°.

Question 7

- Answer: B — 90°
- Working & Reason: Since |ED| = |DC| = |CE|, triangle EDC is equilateral, so all its angles are 60°.

On straight line AED, ∠AEC = 180° − 60° = 120°. Triangle AEC is isosceles (|AE| = |CE|), so base angles are (180° − 120°) / 2 = 30°. On straight line BCD, ∠ACB + ∠ECA + ∠ECD = 180°. x + 30° + 60° = 180° x = 180° − 90° = 90°.

Question 8

- Answer: B — 25 pupils
- Working & Reason: Sum the frequencies (heights) of each score column:

Total pupils = 3 (for mark 2) + 2 (mark 3) + 1 (mark 4) + 4 (mark 5) + 5 (mark 6) + 4 (mark 7) + 1 (mark 8) + 2 (mark 9) + 3 (mark 10) Total = 3 + 2 + 1 + 4 + 5 + 4 + 1 + 2 + 3 = 25 pupils.